STACKS, DIFFERENTIAL GRADED SHEAVES: LEMMA-HOM-DERIVED

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4507-4520 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 4522-4549, the complete author proof (26 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 23 of queue_sdga.json (pb=1219, stmt 4507-4520, proof 4522-4549).

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-hom-derived' also exists in stacks-dga — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 04VF: sources/stacks-sdga/pinned-original-source.tex, source lines 4549–4549. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FT5: sources/stacks-sdga/pinned-original-source.tex, source lines 4458–4470. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FSX: sources/stacks-sdga/pinned-original-source.tex, source lines 4024–4039. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $(\mathcal{A}, d)$ be a sheaf of differential graded algebras on $(\mathcal{C}, \mathcal{O})$. Let $\mathcal{M}$ and $\mathcal{N}$ be differential graded $\mathcal{A}$-modules. Let $\mathcal{N} \to \mathcal{I}$ be a quasi-isomorphism with $\mathcal{I}$ a graded injective and K-injective differential graded $\mathcal{A}$-module. Then*

$$\mathrm{Hom}_{D(\mathcal{A},d)}(\mathcal{M}, \mathcal{N}) = \mathrm{Hom}_{K(Mod(\mathcal{A},d))}(\mathcal{M}, \mathcal{I})$$

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Proof. Since $\mathcal{N} \to \mathcal{I}$ is a quasi-isomorphism we see that

$$\mathrm{Hom}_{D(\mathcal{A},\mathrm{d})}(\mathcal{M}, \mathcal{N}) = \mathrm{Hom}_{D(\mathcal{A},\mathrm{d})}(\mathcal{M}, \mathcal{I})$$

In the discussion preceding Definition 0FT5 we found, using Lemma 0FSX, that any morphism $\mathcal{M} \to \mathcal{I}$ in $D(\mathcal{A}, \mathrm{d})$ can be represented by a morphism $f : \mathcal{M} \to \mathcal{I}$ in $K(\mathit{Mod}(\mathcal{A}, \mathrm{d}))$. Now, if $f, f' : \mathcal{M} \to \mathcal{I}$ are two morphism in $K(\mathit{Mod}(\mathcal{A}, \mathrm{d}))$, then they define the same morphism in $D(\mathcal{A}, \mathrm{d})$ if and only if there exists a quasi-isomorphism $g : \mathcal{I} \to \mathcal{K}$ in $K(\mathit{Mod}(\mathcal{A}, \mathrm{d}))$ such that $g \circ f = g \circ f'$, see Categories, Lemma 04VF. However, by Lemma 0FSX there exists a map $h : \mathcal{K} \to \mathcal{I}$ such that $h \circ g = \mathrm{id}_{\mathcal{I}}$ in in $K(\mathit{Mod}(\mathcal{A}, \mathrm{d}))$. Thus $g \circ f = g \circ f'$ implies $f = f'$ and the proof is complete. $\square$